DISSERTATION PROJECT

FFR Physiological Plausibility Scoring

Using a Conditional GAN to detect out-of-distribution CFD samples from biased stenosis definitions

CFD · TensorFlow · Keras · Python · cGAN · OOD Detection

PROBLEM & OBJECTIVE

Background

The Problem

- Two CFD datasets exist for coronary artery stenosis simulation
- Reference dataset uses diameter-based severity — physiologically correct
- Biased dataset uses area-based severity — produces inconsistent FFR values
- Area-based severity systematically underestimates obstruction
- Biased FFR values cluster unnaturally high (mean 0.93 vs 0.73)

The Objective

- Learn the distribution of physiologically plausible FFR values from reference data only
- Use the trained discriminator to score any new sample's physiological plausibility
- Detect whether biased samples are out-of-distribution
- Output: plausibility score $\in [0, 1]$ per sample
- No supervised learning on the biased dataset — evaluation only

DATA

Dataset Overview

1,066	1,200	130	3
Reference samples (real)	Biased samples (fake)	Removed CFD artefacts (FFR<0)	Input features per sample

Reference Dataset (Real)

Severity definition Diameter reduction

FFR range 0.002 – 0.956

FFR mean / std 0.73 / 0.21

Artery length 100 mm

Used for Training only

Biased Dataset (Fake)

Severity definition Area reduction

FFR range 0.41 – 0.96

FFR mean / std 0.93 / 0.03

Artery length 120 mm

Used for Evaluation only

ARCHITECTURE

cGAN Model Design

Generator (training only)

Noise (16-dim) + Condition (9 features)

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Dense 128 → LeakyReLU → BatchNorm

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Dense 128 → LeakyReLU → BatchNorm

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Dense 64 → LeakyReLU

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Output: Generated FFR ∈ [0,1]

Role: challenges discriminator during training

Discriminator (the scorer)

FFR (1) + Condition (9 features) = 10 inputs

↓

Dense 256 → LeakyReLU → Dropout 30%

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Dense 128 → LeakyReLU → Dropout 20%

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Dense 64 → Dense 32 → LeakyReLU

↓

Output: Plausibility Score ∈ [0,1]

Role: the final plausibility scorer used for evaluation

METHOD

Training Strategy

1 Feature Engineering

Added 3 derived features: expected_ffr, ffr_residual, ffr × severity. These encode the physiological FFR–severity relationship directly, giving the discriminator detectable signal.

2 Unsupervised Warm-up

80 epochs using only internal shuffling — no labels, no fake data. Shuffled FFR and shuffled conditions create mismatched pairs as negatives. Anchors D(real)=0.814 before adversarial training.

3 Adversarial Training

400 epochs. Discriminator trains 2× per generator step. Real score penalty added: penalises discriminator when real scores fall below 0.95. D LR < G LR to prevent collapse.

4 OOD Evaluation

Discriminator scored on held-out real test set and entire fake dataset. Threshold set at 10th percentile of real test scores. No fake data ever used in training.

RESULTS

Evaluation Metrics

0.838	0.106	0.732	0.9963
Real mean plausibility score	Fake mean plausibility score	Separation gap	AUC-ROC

Classification Metrics (10th pct threshold)

Accuracy	98.6%
Sensitivity	90.0%
Specificity	99.75%
Precision	97.96%
F1 Score	0.938
AUC-PR	0.9791
KS Statistic	0.9367

OOD Detection Performance

Fake below 10th pct	99.9%
Fake below 25th pct	99.9%
Real retained (10th)	90.0%
Fake wrongly kept	1 / 1200

p-value (KS test): 1.05×10^{-149} — distributions are statistically distinct

CONCLUSION

Key Findings

- ✓ cGAN discriminator learned the physiological distribution from reference data only — no fake data used in training
- ✓ Real samples scored mean 0.838 · Fake samples scored mean 0.106 · Separation gap of 0.732
- ✓ 99.9% of biased samples detected as out-of-distribution at the 10th percentile threshold
- ✓ AUC-ROC 0.9963 · AUC-PR 0.979 · KS statistic 0.937 · Accuracy 98.6%

Tools: Python · TensorFlow / Keras · Scikit-learn · Google Colab · Matplotlib · NumPy · Pandas